

Problem Set 4: Linear and Non-Linear Equations

Readings: Newman Ch. 6

Due: Wednesday, October 7th 11:59 PM

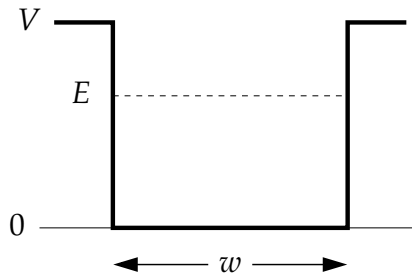
Instructions: Work through each problem, following the instructions for the requirements of the code and their outputs. Some starting code is provided for each problem on the course website. Problem sets should be submitted in one batch, with each problem uploaded as it's own python script (.py file) on the HuskyCT assignment page with the filename: ps[PROBLEM SET NUMBER]_[QUESTION NUMBER]_[LASTNAME].py.

Parameters:

- (1) Submitted solutions are restricted to imports of external functions from standard built-in Python libraries (i.e. math, time, os, sys, etc.). In addition, imports from matplotlib.pyplot and numpy are permitted for basic array operations.
- (2) All plots should have both axes clearly labeled and include a title. Plots should include either legends or annotations for differentiating plot elements, when applicable.

1. SQUARE WELL ENERGY

Consider a square potential well of width w , with walls of height V :



Using Schrödinger's equation, it can be shown that the allowed energies E of a single quantum particle of mass m trapped in the well are solutions of the following equations:

$$\tan \sqrt{w^2 m E / 2 \hbar^2} = \begin{cases} \sqrt{(V - E) / E} & \text{for the even numbered states,} \\ -\sqrt{E / (V - E)} & \text{for the odd numbered states,} \end{cases}$$

where the states are numbered starting from 0, with the ground state being state 0, the first excited state being state 1, and so forth.

Write a program to calculate the energy levels of a square well for an electron (mass 9.1094×10^{-31} kg) in a well with $V = 20$ eV and $w = 1$ nm using a binary search method. The program should:

- a. Plot the three quantities:

$$y_1 = \tan \sqrt{w^2 m E / 2 \hbar^2}, \quad y_2 = \sqrt{\frac{V - E}{E}}, \quad y_3 = -\sqrt{\frac{E}{V - E}},$$

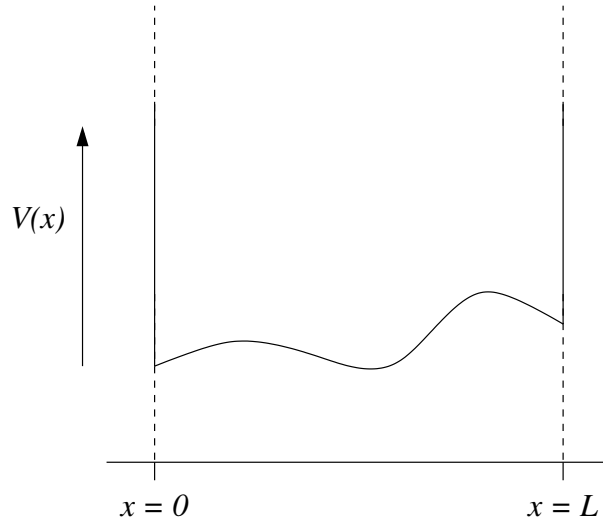
on the same panel, as a function of E from $E = 0$ eV to $E = 20$ eV.

- b. Have a list of the interval bounds to search for the energies of the first six (6) energy levels of the particle. (The plot is helpful here).

- c. Include a function to calculate the values of an energy level to an accuracy of 0.001 eV, given some initial guess, using binary search.
- d. Print and calculate the first six (6) energy levels of the well in electron volts and mark their locations on the plot from (a.).
- e. Save the plot as `squarewell_[LASTNAME].png`.

2. ASYMMETRIC QUANTUM WELL

Suppose, that rather than a square well in which the potential $V(x)$ is constant within the well, take, instead, that $V(x)$ is a varying function of x within the well.



For arbitrary functions of x , there is no guarantee of an analytic solution, which is why they are often solved numerically. In a pure state of energy E , the spatial part of the wavefunction obeys the time-independent Schrödinger equation $\hat{H}\psi(x) = E\psi(x)$, where the Hamiltonian operator $\hat{H}$ is given by

$$\hat{H} = -\frac{\hbar^2}{2M} \frac{d^2}{dx^2} + V(x).$$

For simplicity, let's assume that the walls of the well are infinitely high, so that the wavefunction is zero outside the well, which means it must *go to zero* at $x = 0$ and $x = L$. In that case, the wavefunction can be expressed as a Fourier sine series thus:

$$\psi(x) = \sum_{n=1}^{\infty} \psi_n \sin \frac{\pi n x}{L},$$

where $\psi_1, \psi_2, \dots$ are the Fourier coefficients. Noting that, for m, n positive integers

$$\int_0^L \sin \frac{\pi m x}{L} \sin \frac{\pi n x}{L} dx = \begin{cases} L/2 & \text{if } m = n, \\ 0 & \text{otherwise,} \end{cases}$$

it can be shown that the Schrödinger equation $\hat{H}\psi = E\psi$ implies that

$$\sum_{n=1}^{\infty} \psi_n \int_0^L \sin \frac{\pi m x}{L} \hat{H} \sin \frac{\pi n x}{L} dx = \frac{1}{2} L E \psi_m.$$

Hence, defining a matrix $\mathbf{H}$ with elements

$$\begin{aligned} H_{mn} &= \frac{2}{L} \int_0^L \sin \frac{\pi m x}{L} \hat{H} \sin \frac{\pi n x}{L} dx \\ &= \frac{2}{L} \int_0^L \sin \frac{\pi m x}{L} \left[-\frac{\hbar^2}{2M} \frac{d^2}{dx^2} + V(x) \right] \sin \frac{\pi n x}{L} dx, \end{aligned}$$

the Schrödinger's equation can be written in matrix form as $\mathbf{H}\boldsymbol{\psi} = E\boldsymbol{\psi}$, where $\boldsymbol{\psi}$ is the vector $(\psi_1, \psi_2, \dots)$. Thus $\boldsymbol{\psi}$ is an eigenvector of the Hamiltonian matrix $\mathbf{H}$ with eigenvalue E . If we can calculate the eigenvalues of this matrix, then we know the allowed energies of the particle in the well.

For the case $V(x) = ax/L$, the integral in H_{mn} can be evaluated analytically to find a general expression for the matrix elements H_{mn} (where the matrix is real and symmetric):

$$H_{mn} = \begin{cases} 0 & \text{if } m \neq n \text{ and both even or both odd,} \\ -\frac{8amn}{\pi^2(m^2 - n^2)^2} & \text{if } m \neq n \text{ and one is even, one is odd,} \\ \frac{a}{2} + \frac{\pi^2 \hbar^2 m^2}{2ML^2} & \text{if } m = n. \end{cases}$$

Write a program to evaluate H_{mn} when the particle in the well is an electron, the well has width $5 \text{ \AA}$, and $a = 10 \text{ eV}$. (The mass and charge of an electron are $9.1094 \times 10^{-31} \text{ kg}$ and $1.6022 \times 10^{-19} \text{ C}$ respectively.) The program should:

- Include a function to evaluate your expression for H_{mn} for arbitrary m and n .
- Use the function to generate a 10×10 array of the elements of $\mathbf{H}$ up to $m, n = 10$.
- Calculate the eigenvalues of this matrix using the appropriate function from `numpy.linalg` and hence print out, in units of electron volts, the first ten energy levels of the quantum well, within this approximation.
- Repeat steps b. and c. but for a 100×100 array for $\mathbf{H}$ instead. Note the convergence behavior for the calculation.
- Calculate the wavefunction $\psi(x)$ for the ground state and the first two excited states of the well.
- Make a plot with three curves showing the probability density $|\psi(x)|^2$ as a function of x in nm for each of the three states.
- Save the figure as `wellpdf_[LASTNAME].png`.

Recall that wavefunctions must be normalized to satisfy the condition: $\int_0^L |\psi(x)|^2 dx = 1$.

3. OVERRELAXATION

The relaxation method involves iterating the equation $x_{i+1} = f(x_i)$, starting from an initial guess x_0 , until it converges. However, it is possible in some cases to make the method work even faster using the technique of *overrelaxation*. In the overrelaxation method, if x_{i+1} is increasing, then we can deliberately overshoot the calculated value, in the hope that this will get us closer to the final solution faster.

The overrelaxation method involves iteration of the modified equation:

$$x_{i+1} = x_i + (1 + \omega) \Delta x,$$

where $\Delta x = x_{i+1} - x_i = f(x_i) - x_i$ and ω is an overshooting parameter. If the parameter ω is zero, then this expression reduces to that of the ordinary relaxation method. For $\omega > 0$ the method takes the amount Δx by which the value of x_i would have been changed and changes it by a little more. Substituting for Δx , we can write the modified equation in its typical form:

$$x_{i+1} = x_i + (1 + \omega) [f(x_i) - x_i] = (1 + \omega)f(x_i) - \omega x_i,$$

It can be shown that the error on x_{i+1} is given by

$$\epsilon_{i+1} \simeq \frac{x_i - x_{i+1}}{1 - 1/[(1 + \omega)f'(x_i) - \omega]}.$$

Thus, we can see that the choice of the overshooting parameter will influence the convergence behavior.

Consider an equation of the form

$$x = 1 - e^{-cx},$$

where c is a known parameter and x is the unknown variable. This type of equation arises in a variety of situations: including the physics of contact processes, mathematical models of epidemics, and the theory of random graphs. As we've discussed in class, this equation is solvable by the relaxation method.

Write a program to solve the above equation using the (over)relaxation method. The program should:

- Include a function to solve an equation given the function $f(x)$, the overshooting parameter ω , to a chosen accuracy level. The function should print the number of iterations required for the solution to converge.
- Use the function to solve for x using the original relaxation method ($\omega = 0$) for the case $c = 2$ to an accuracy of at least 10^{-6} .
- Use the function to solve for x using the *overrelaxation* method for the case $c = 2$ to an accuracy of at least 10^{-6} , by optimizing to find the ω value required to arrive to a solution within less than half of the iterations as in part a.
- Plot the results of your relaxation function (with $\omega = 0$) for x as function of the parameter c , for values of c between 0 to 3. You should see a clear transition from a regime in which $x = 0$ to a regime of nonzero x . In physics this phase transition is known as the *percolation transition*; in epidemiology it is the *epidemic threshold*.
- Save this plot as `relax_[LASTNAME].png`.

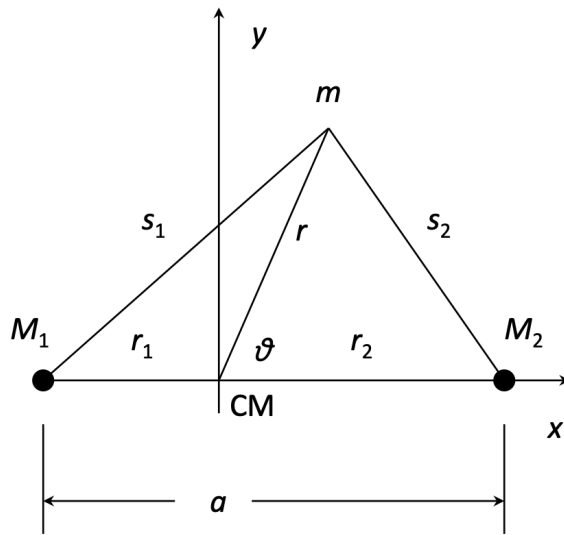
Hint: You will want to experiment with values of ω between 0 and 1

4. LAGRANGE POINT

In celestial mechanics, Lagrange points are equilibrium points where a test mass can remain stationary relative to two larger celestial bodies. The locations of Lagrange points are solutions to the circular restricted three-body problem in which the effective potential surface in the $x - y$ plane of all three orbits is described by:

$$\Phi(r) = -G \left(\frac{M_1}{s_1} + \frac{M_2}{s_2} \right) - \frac{G(M_1 + M_2)}{2a^3} r^2$$

where M_1 and M_2 are the masses of the two bodies, a is the semi-major axis, the total distance between the two masses, and s_1 and s_2 are the separations between the bodies and the position of the satellite at some position r relative to the origin, defined at the location of the center of mass between M_1 and M_2 . The first term is from the gravitational potential exerted on the test mass from the two bodies, while the second term accounts for the centrifugal acceleration of the rotating frame with angular velocity given by Kepler's third law: $\omega = \sqrt{G(M_1 + M_2)/a^3}$



The locations of the Lagrange points are then at the extrema of the potential surface, that is where $\nabla\Phi = 0$. Consider the Earth-Moon system for which there are 5 Lagrange points, where L_1 , L_2 , and L_3 are co-linear, lying on the same line as the separation vector between the Earth M_1 and the Moon M_2 (the x -axis in the plot above). L_4 and L_5 lie along the orbit of the Moon, leading and trailing by some angle θ .

Write a program to calculate the 2D potential surface of the Earth-Moon system and use it to determine the locations of L_1 Lagrange point. The program should:

- Define a set of coordinates in x and y in units of the semi-major axis a with the origin set to be the center of mass of the Earth-Moon system.
- Include a function to calculate $\Phi(x, y)$ on a grid of points from $-1.5a$ to $1.5a$ in both dimensions in units of $-3GM_1/2a$.
- Make a 2D density plot of the potential vs. x and y . Overplot loci of equipotential using `matplotlib.contour` to visualize the shape of the

Hint: Consider how you can scale your result to the desired units by assuming $G = M_1 = a = 1$

potential surface. The locations of the co-linear Lagrange points will be saddle points on the surface, while L_4 and L_5 will appear as extrema.

- d. Save your plot as Lagrange_`[LASTNAME]`.png.
- e. Define an initial guess for the coordinates of L_1 , the Lagrange point between the Earth and the Moon based on your plot.
- f. Include a function to solve for the location of an extrema in Φ using the gradient descent method adaptation of the secant method. Note that for co-linear Lagrange points, the problem reduces down to an equation of a single variable $\Phi(x, y = 0)$.
- g. Calculate and print the distance of the L_1 Lagrange point *from the center of the Earth* in units of km to the third decimal point. With well-informed initial guesses and appropriate choice of γ , your code should be able to reproduce the known value $L_1 = 3.844 \times 10^5$ km in less than 50 iterations.